MIL-C-17903C(SHIPS)
1 April 1957
SUPERSEDING
MIL-C-17903B(SHIPS)
17 February 1956

MILITARY SPECIFICATION

CONTROL, ELECTRO-HYDRAULIC STEERING GEAR

SYNCHRO HAND ELECTRIC

1. SCOPE

1.1 This specification covers the design, manufacture and test of synchro hand electric steering control for electro-hydraulic steering gears employing variable delivery pumps equipped with hydraulic servo-control.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shockproof Equipment, Class HI (High Impact) Shipboard Application Tests for.
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-T-17113 - Tests; Shock, Vibration, and Inclination (for Electronic Equipment) General Specification.
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

FSC 2030

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NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-2 - Drawing Sizes.

DRAWINGS

BUREAU OF ORDNANCE
561794 - Synchro Units Size 7 - Requisition Drawing Outline.

BUREAU OF SHIPS
S6505-1568789 - Synchro Hand Electric Control for Steering Gear - General Arrangement for Console Mounting.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 In order to minimize the corrosive effect due to electrolytic action between dissimilar metals, contact between dissimilar metals shall be limited to those metals having a small difference of electrolytic potential in the electrochemical series. Contact between aluminum alloys and other metals shall be avoided as much as practicable. Where practicable, welded or brazed construction shall be used in lieu of castings. No cast iron shall be used.

3.1.2 When required (see 6.1), all material shall have low magnetic permeability, except certain parts of the electrical equipment and components, such as anti-friction bearings, the manufacture of which would involve a disproportionate amount of time and money if made of nonmagnetic materials. Other small components may be constructed of magnetic material if no suitable nonmagnetic substitute can be utilized. Aluminum alloys, brass, bronze, copper-nickel alloys, nonmagnetic corrosion-resisting steel, and nickel-copper alloys should be used in lieu of magnetic metals and alloys. The use of other nonmagnetic materials will be considered but shall be submitted to the bureau or agency concerned for final approval.

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3.1.3 Nonmagnetic materials shall have a permeability of less than 2.0 after fabrication.

3.1.4 All material shall be suitable in all respects for the purpose intended, and shall be free from defects which might affect the serviceability of the steering gear control.

3.2 Design features. -

3.2.1 General. - The synchro control system shall provide hand-electric steering control of a Naval vessel by means of a steering wheel driving a rudder order synchro control transmitter unit which controls a synchro receiver in the steering engine room as specified (see 6.1). The output shaft of each receiver is to transmit to the steering gear a displacement proportional to the angular displacement of the steering wheel.

3.2.2 Shock and vibration. - All components of the steering control shall be designed and constructed to withstand HI shock tests in accordance with Specification MIL-S-901 and vibration tests in accordance with Specification MIL-T-17113 (see 4.2.2).

3.2.3 Cables. - Electric wires and cables and the application shall be in accordance with Specification MIL-C-915.

3.2.4 Power supply. - The equipment shall operate from a 120-volt, single phase, 60 cycle alternating current (a.c.) power supply.

3.2.5 Coordination with other equipment. - The synchro control transmitter unit shall be suitable for mounting in a steering control console.

3.2.6 Control system characteristics. - The synchro control system shall have the following characteristics:

(a) The steering wheel shall rotate not more than 12 turns for hard-over to hard-over rudder movement of the steering gear which it is controlling.
(b) Damping by means of inertia damper shall be provided if required at the receiver to prevent pulling the steering control synchros out of step and to prevent motoring.
(c) The synchro control system shall be capable of developing a maximum static torque equivalent to 300 percent of the connected load without exceeding 36 degrees displacement of synchros.
(d) Steering transmitter and receiver synchros shall be size 7 in accordance with Drawing 561794.

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3.2.7 Components. -

3.2.7.1 Synchro control transmitter unit. - The synchro control transmitter unit shall consist of one or more size 7 single phase Navy synchros (Navy stock number 7HG-Z17-M-61447-7420; MK 9 Mod 2, MK 9 Mod 3, or MK 9 Mod 6) as specified (see 6.1), a steering wheel, and gearing mounted on a panel suitable for installation on a console furnished by others. The unit shall provide for connection of an indicating light furnished by others, to indicate control power available. Synchros with 3 phase stators will not be acceptable.

3.2.7.1.1 The mounting of all synchros shall permit turning of the stator to set the electrical zero.

3.2.7.1.1.1 The steering wheel shall be of the smooth rim, disc type, 20 inches outside diameter. The wheel shaft shall be not less than 1 inch in diameter. Sufficient inertia shall be added to the wheel to prevent pulling the synchros out of step by rapid acceleration or reversal of the wheel.

3.2.7.1.1.2 The unit for mounting on the steering control console shall be watertight. The panel shall not be drilled for mounting bolts.

3.2.7.1.1.3 The noise level of the synchro transmitter shall not exceed 40 decibels with synchros displaced from 0 to 36 degrees.

3.2.7.1.1.4 The dimension from the center line of the wheel shaft to the top of the mounting panel shall not exceed 3 inches.

3.2.7.1.2 Overall dimensions of the synchro transmitter shall be limited to those shown on Drawing S6505-1568789.

3.2.7.2 Synchro receiver. - Two synchro receivers shall be provided unless otherwise specified in the contract or order. The mounting for the synchro receiver shall permit turning of the stator to set the electrical zero. The synchro receiver shall be capable of being overhauled in either direction of rotation if torque is applied to its output shaft.

3.3 Drawings. -

3.3.1 Final drawings. - One complete set of reproducible drawings, approved by the bureau or agency concerned, showing assemblies, subassemblies and wiring diagrams with certification data shall be furnished in accordance with type I of Specification MIL-D-963 and Standard MIL-STD-2.

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3.3.2 Drawings for approval. - Manufacturing drawings in triplicate shall be submitted to the bureau or agency concerned for approval when required (see 6.1). The drawings shall show outline dimensions, complete details and bills of material of all system components. In addition, a system schematic gearing diagram shall be furnished indicating gear ratios, pitch diameter of gears, the number of teeth of each gear and method of securing gears to their shafts. An installation drawing, including all system components, showing weight, mounting, and space dimensions of each component shall be furnished. The drawing shall indicate the minimum size and number of wires required between components. Also, a wiring diagram of the complete steering control system including a system schematic and a description of operation shall be furnished. After approval of the manufacturing drawings, the manufacturer will be authorized to procure material and proceed with manufacture in advance of final drawing approval.

3.4 Technical manuals. - Technical manual inserts for inclusion in steering gear technical manuals shall be type B in accordance with Specification MIL-T-15071. Preliminary technical manual inserts, including reproductions of final drawings, shall be approved by the bureau or agency concerned.

3.5 Workmanship. - The workmanship shall be first class in every respect and free from defects which might affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. - The inspection shall be made at the place of manufacture by Government inspectors.

4.2 Test. -

4.2.1 Procedures. - The synchro steering control shall be tested with its associated steering gear to demonstrate that the synchros cannot be pulled out of synchronization by rapid reversing, jerking, or oscillating the steering wheel when 105 volts is impressed on the receiver primary and the leads connecting the synchros have an impedance of not less than 1.3 ohms.

4.2.2 Shock and vibration test. - The contractor will not be required to perform these tests or to submit additional equipment for these tests; however, the Government may perform any tests in accordance with Specifications MIL-S-901 and MIL-T-17113, after delivery, (see 3.2.2). If tests are conducted and the equipment fails, the contractor will be held responsible for modifying the equipment to eliminate the failures.

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4.3 Additional inspection. - Where other specifications form a part of this specification, unless otherwise specified in the contract or order, sampling, examination, and tests shall be conducted as required in the referenced specification.

4.4 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - The equipment, accessories, technical publications, equipment and stock repair parts shall be preserved and packaged by level A or C; packed by level A, B or C (see 6.1) and marked as specified in accordance with Specification MIL-P-17555. The control transmitter unit and the receiver unit shall be individually packaged by method II and the steering wheel, if detached, shall be packaged by method III for level A preservation and packaging.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether nonmagnetic material is required in the synchro control transmitter unit; whether nonmagnetic material is required in the synchro receiver unit (see 3.1.2).
(c) Number of receivers required (see 3.2.1).
(d) Number of size 7 synchros required in the synchro control transmitter unit (see 3.2.7.1).
(e) Whether manufacturing drawings are required (see 3.3.2).
(f) Selection of applicable level of preservation and packaging and packing (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8273

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